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Google Cloud

So, what's the cloud anyway?

Module 1 - Cloud basics
Google Cloud Computing Foundations

Course map

01 — So, what's the cloud anyway?

02 — Start with a solid platform

03 — Use Google Cloud to build your apps

04 — Where do I store this stuff?

05 — There's an API for that!

06 — You can't secure the cloud, right?

07 — It helps to network

08 — Keeping an eye on things

09 — You have the data, but what are you doing with it?

10 — Let machines do the work

Google Cloud skill badges

Objectives

01 Explore cloud computing.

02 Compare and contrast physical, virtual, and cloud architectures.

03 Differentiate IaaS, PaaS, and SaaS.

04 Be introduced to Google Cloud compute, storage, big data, and ML services.

05 Examine the Google network and how it powers cloud computing.

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02 Cloud vs. traditional architecture

03 IaaS, PaaS, and SaaS

04 Google Cloud architecture

05 Quiz

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What is cloud computing?

US National Institute of Standards and Technology

Cloud computing

Cloud computing is a way of using information technology that has these five equally important traits.

01 Customers get computing resources that are on-demand and self-service

02 Customers get access to those resources over the internet, from anywhere

03 The provider of those resources allocates them to users out of that pool

04 The resources are elastic–which means they can increase or decrease as needed

05 Customers pay only for what they use, or reserve as they go

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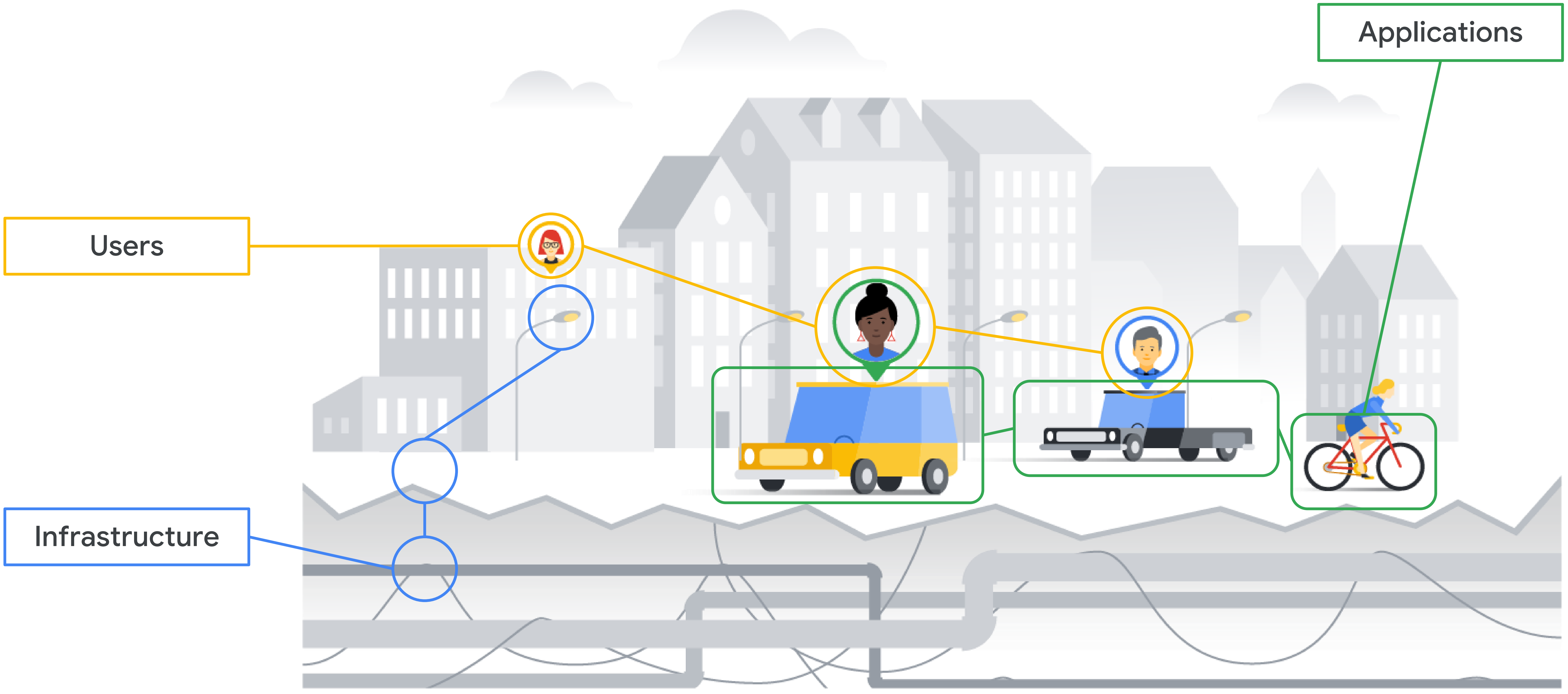
IT infrastructure is like city infrastructure



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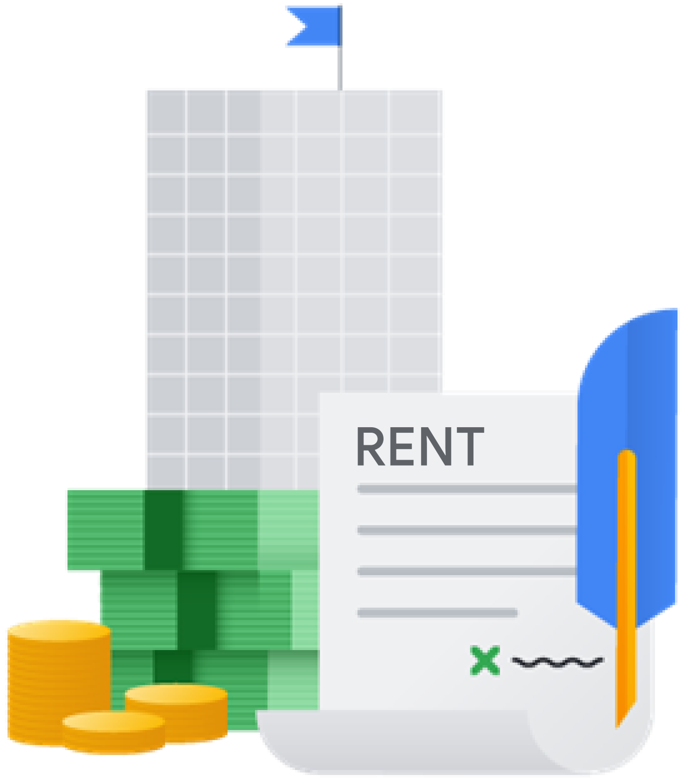
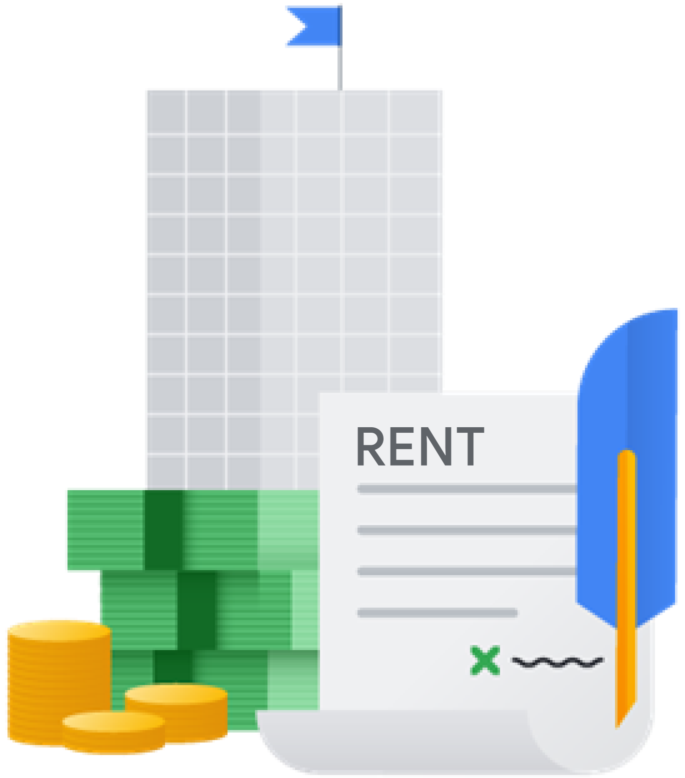
The history of cloud computing



First wave

Colocation

The history of cloud computing



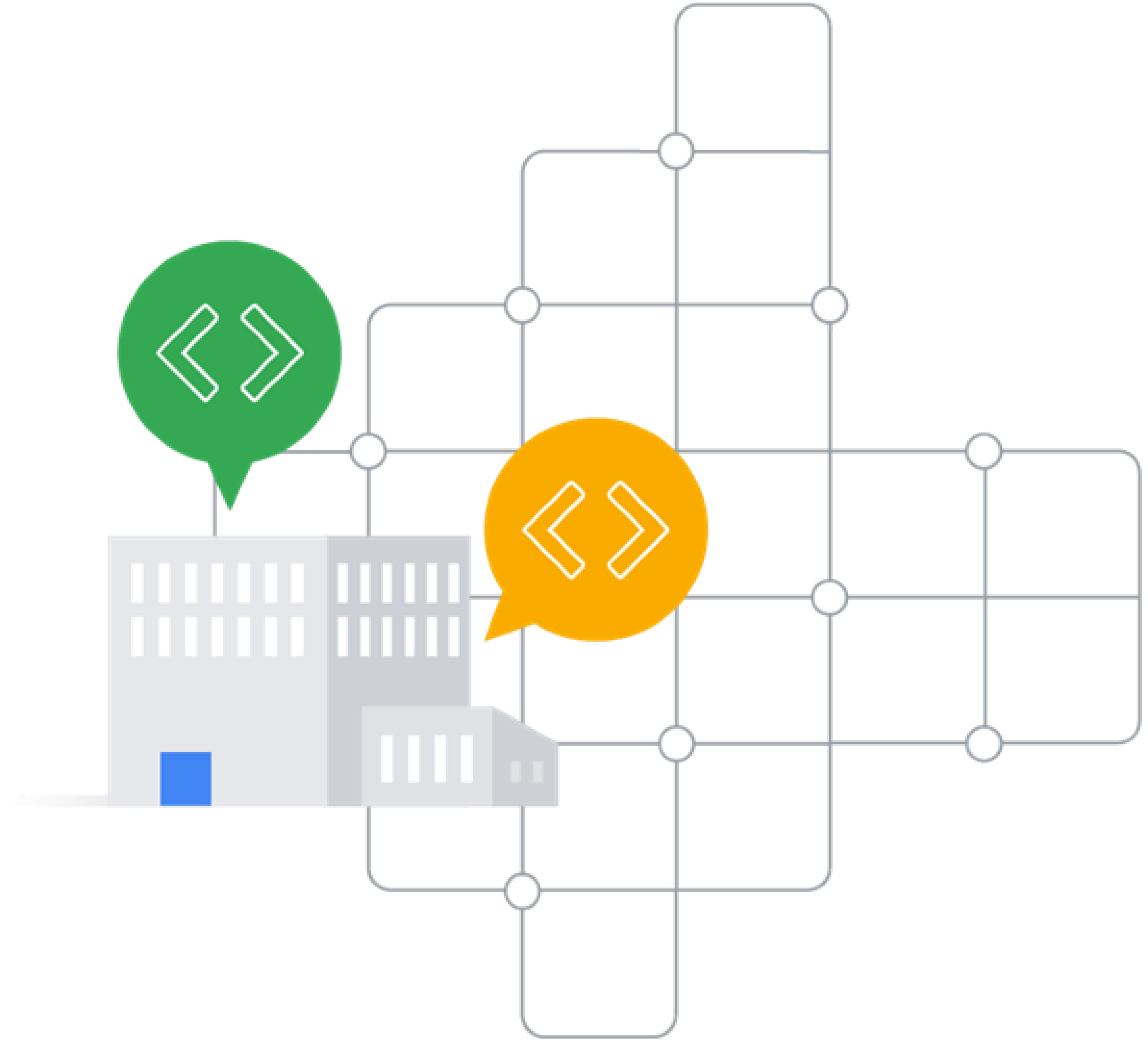
First wave

Colocation

Second wave

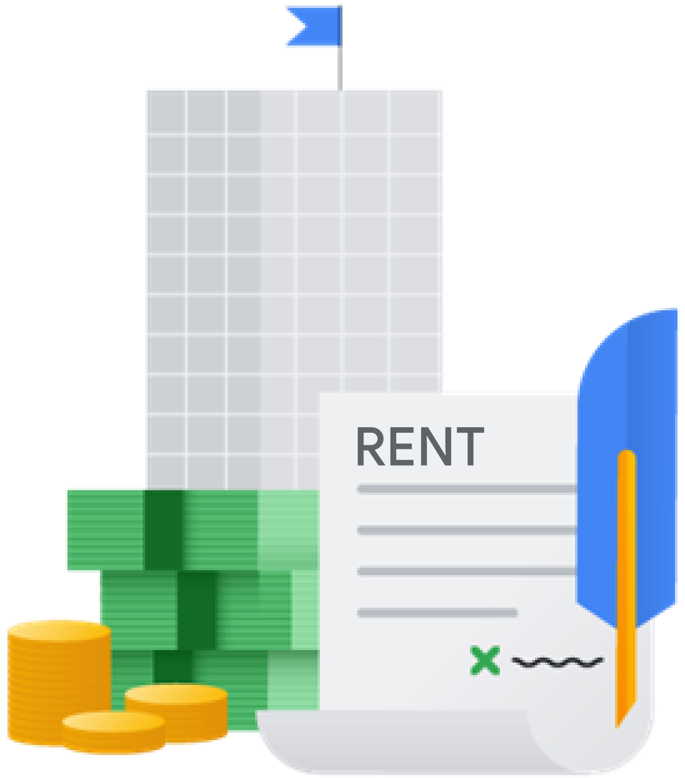
Virtualized
data center

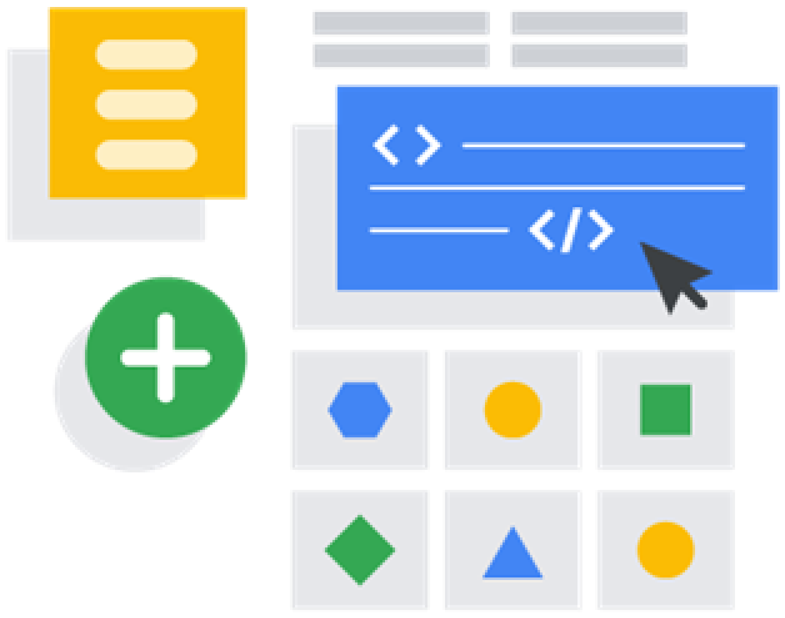
Enterprises still maintain the infrastructure



- User-controlled environment
- User-configured environment

The history of cloud computing





First wave

Colocation

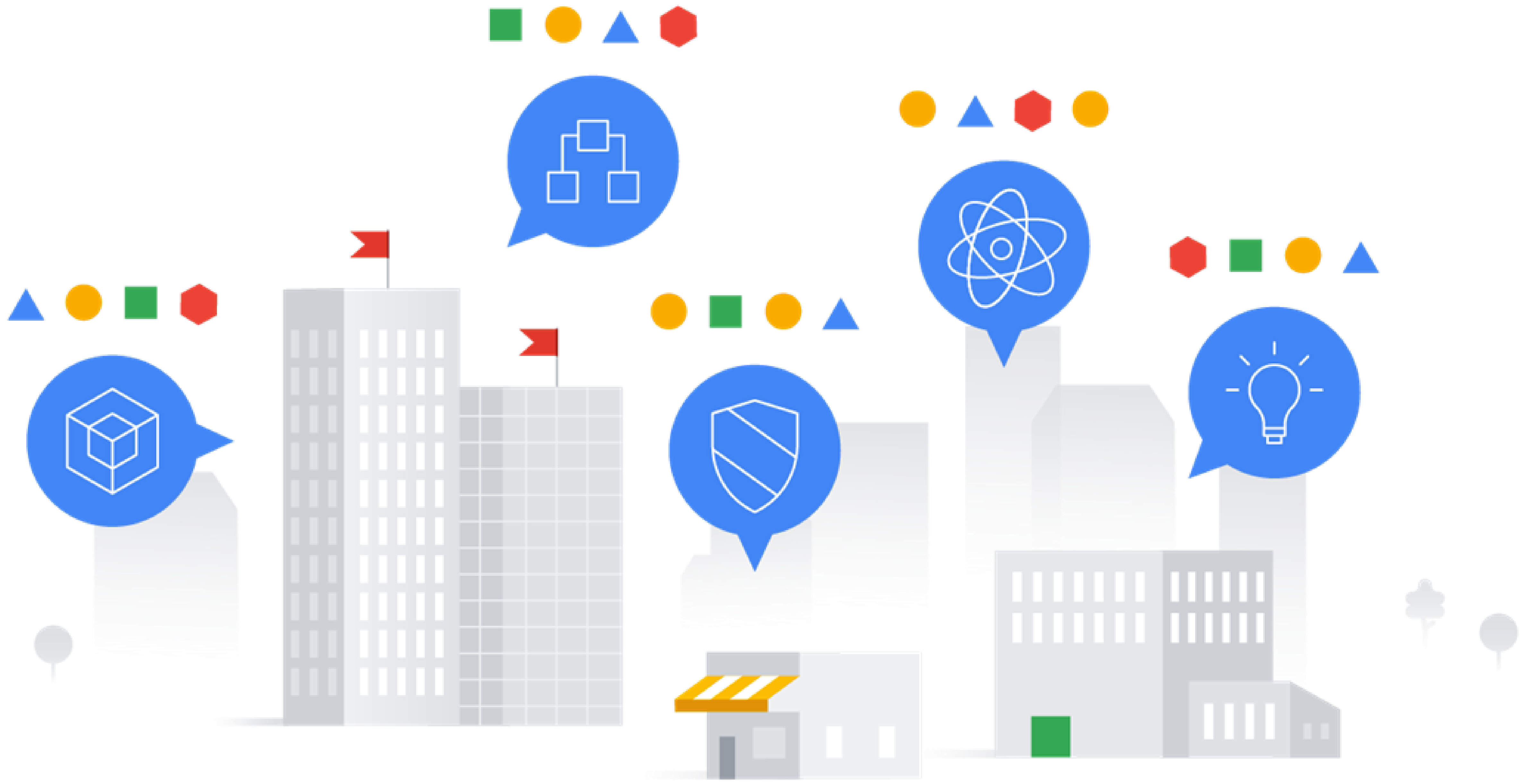
Second wave

Virtualized data center

Third wave

Container-based architecture

Third-wave cloud is available to Google customers

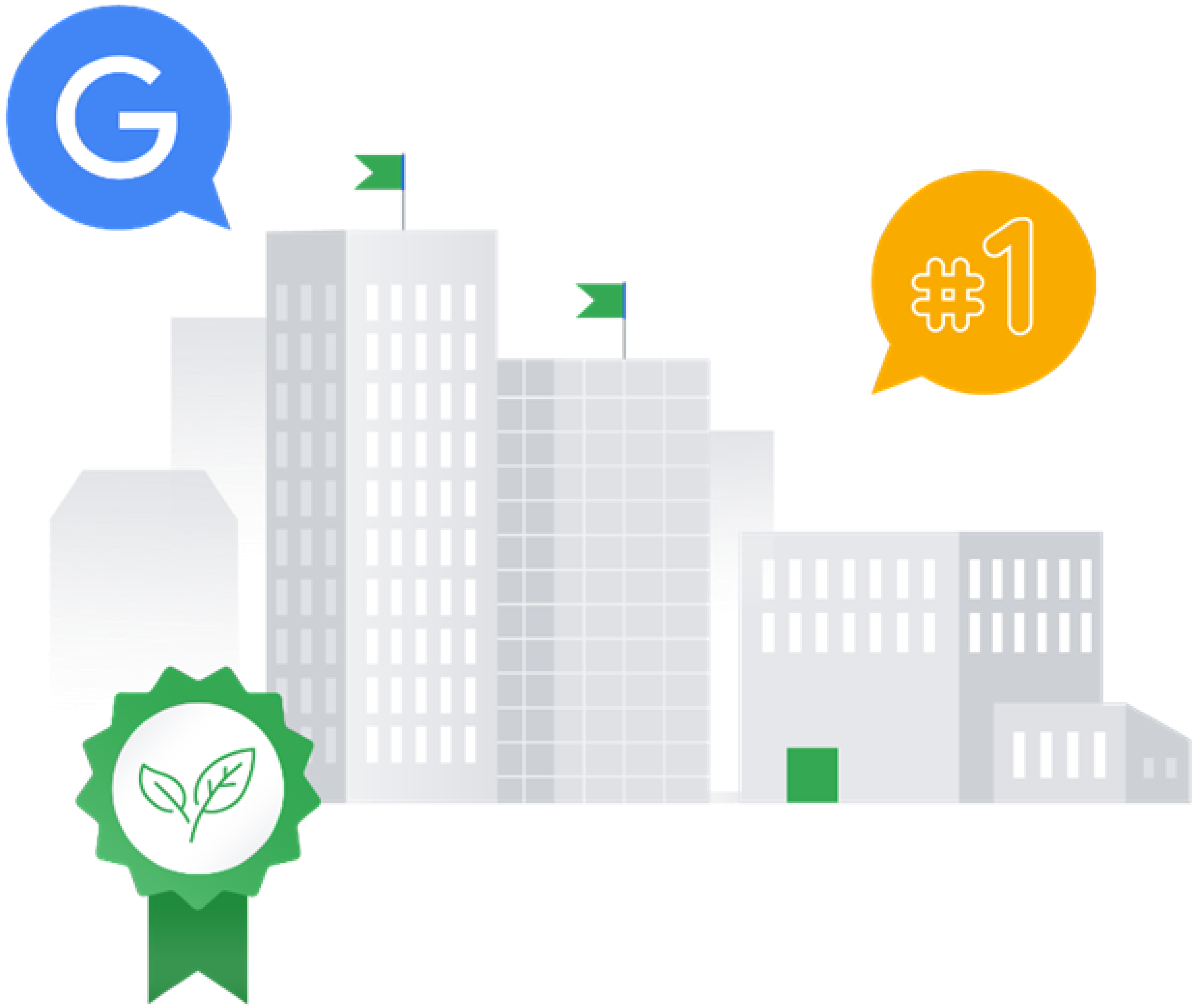


Data center energy consumption

2%

of the world's electricity

Google aims to improve efficiency and reduce waste

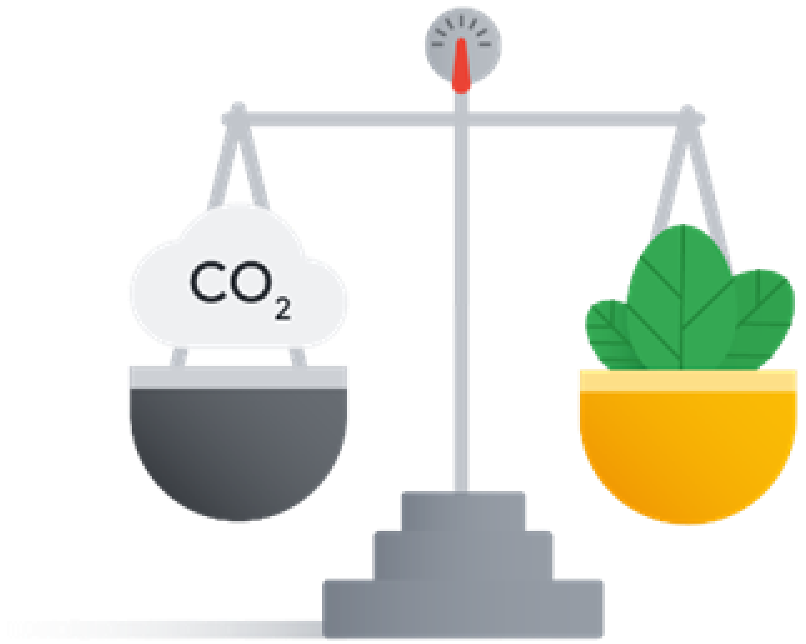


Google's data centers were the first to achieve ISO 14001 certification

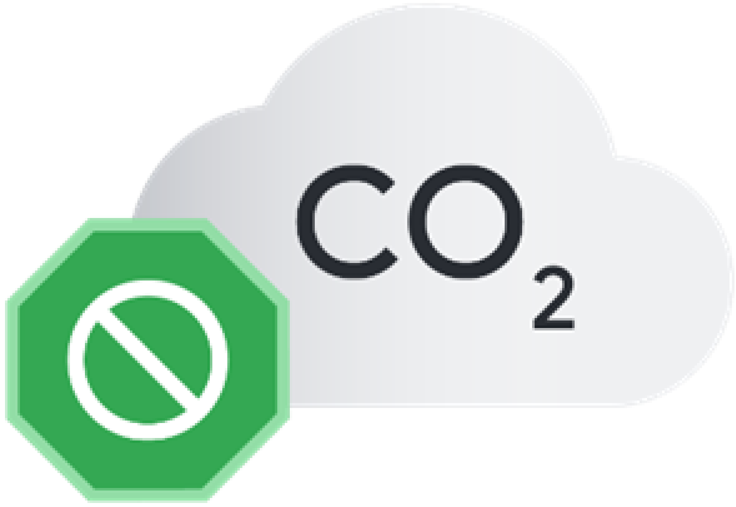
The data center cooling system in Finland is the first of its kind anywhere in the world.

Google's data center, Hamina, Finland

Google's commitment to sustainability



100%



Founding decade	Second decade	2030
Carbon neutral	Renewable energy	Carbon free

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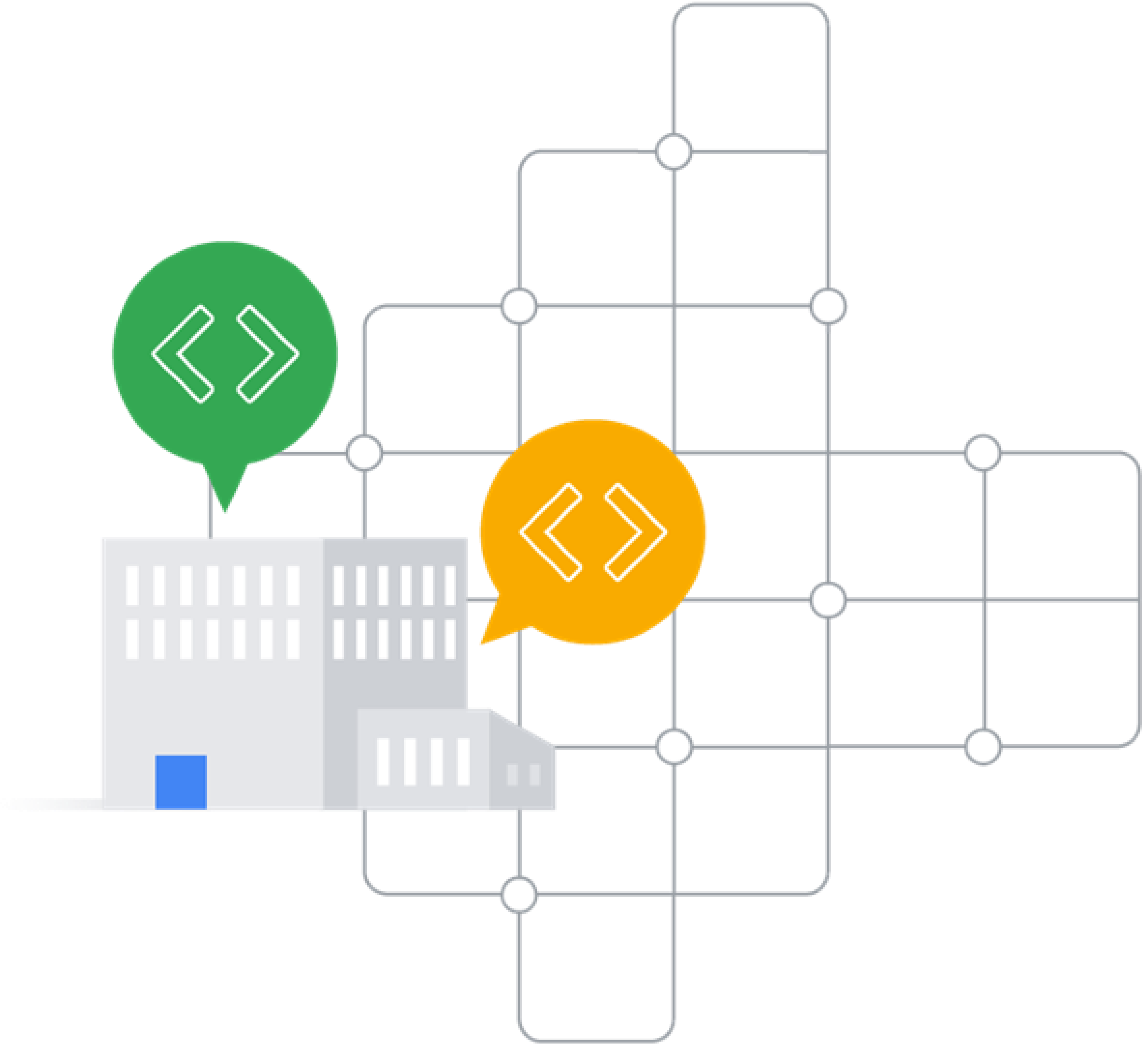
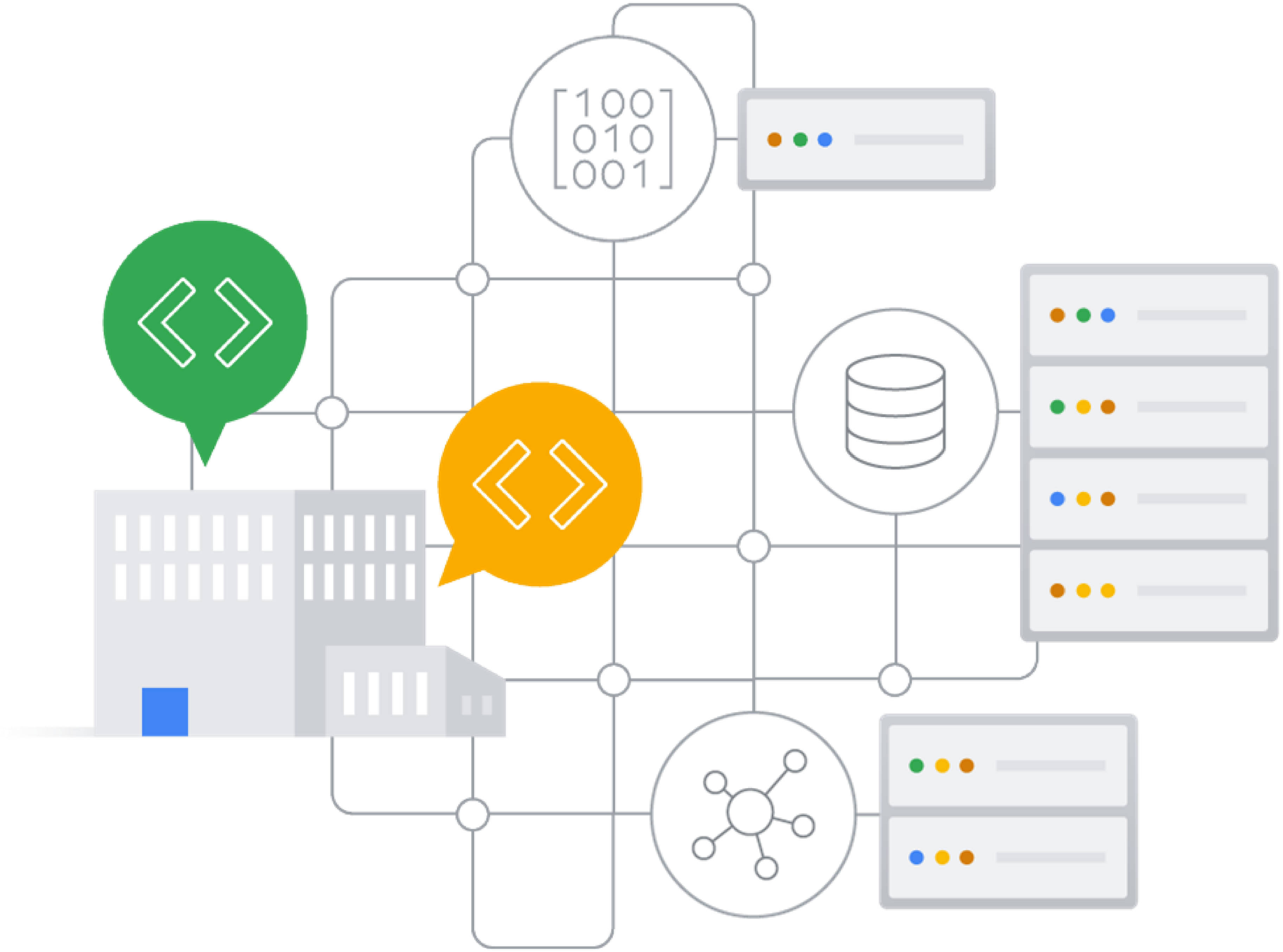
Cloud service offerings



- IaaS - Infrastructure as a service
- PaaS - Platform as a service

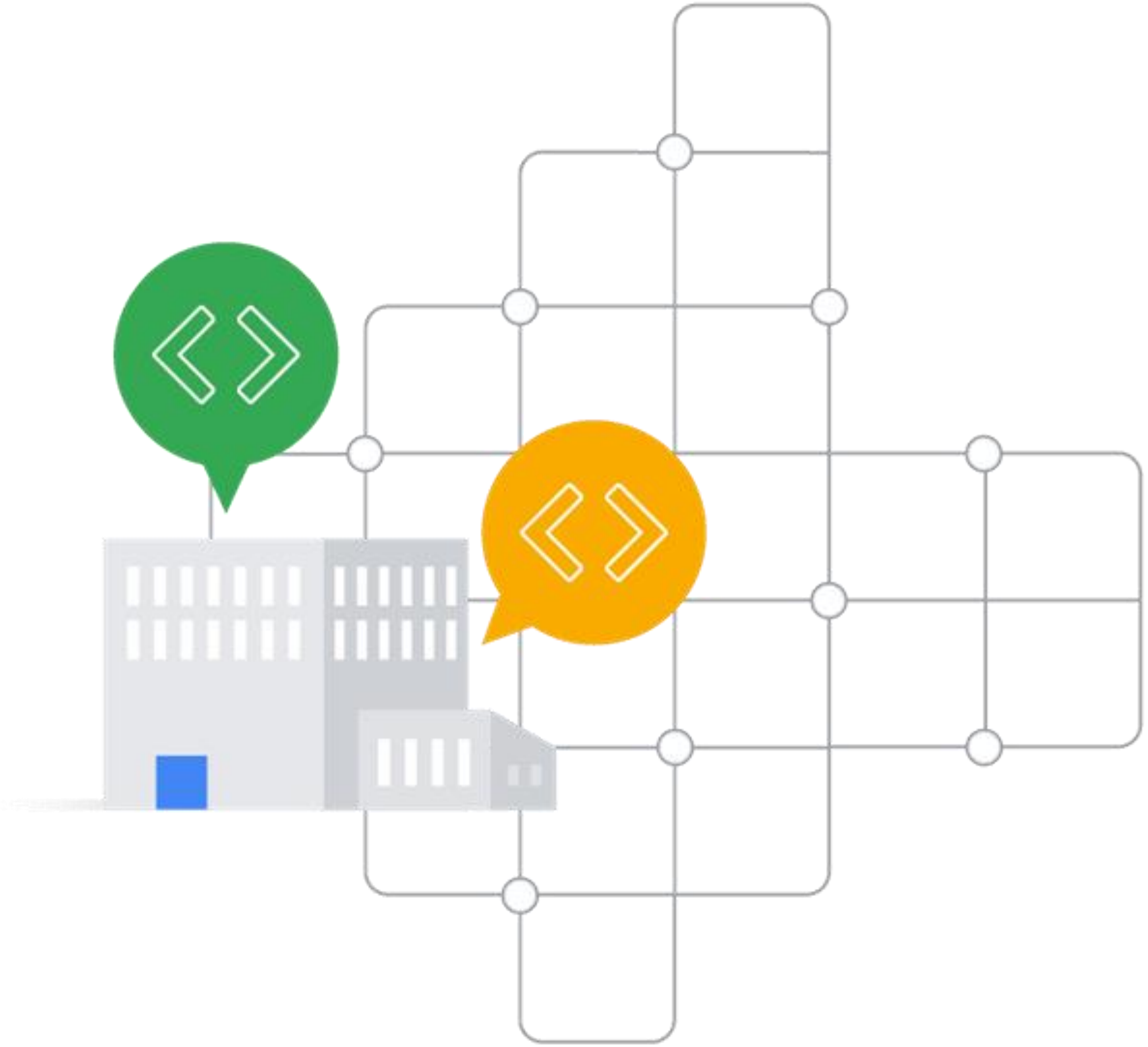
Infrastructure as a Service (IaaS)

- Raw compute
- Storage
- Network capabilities



Platform as a Service (PaaS)

Payment models



IaaS	PaaS
Pay for what they allocate	Pay for what they use

The evolution of cloud computing

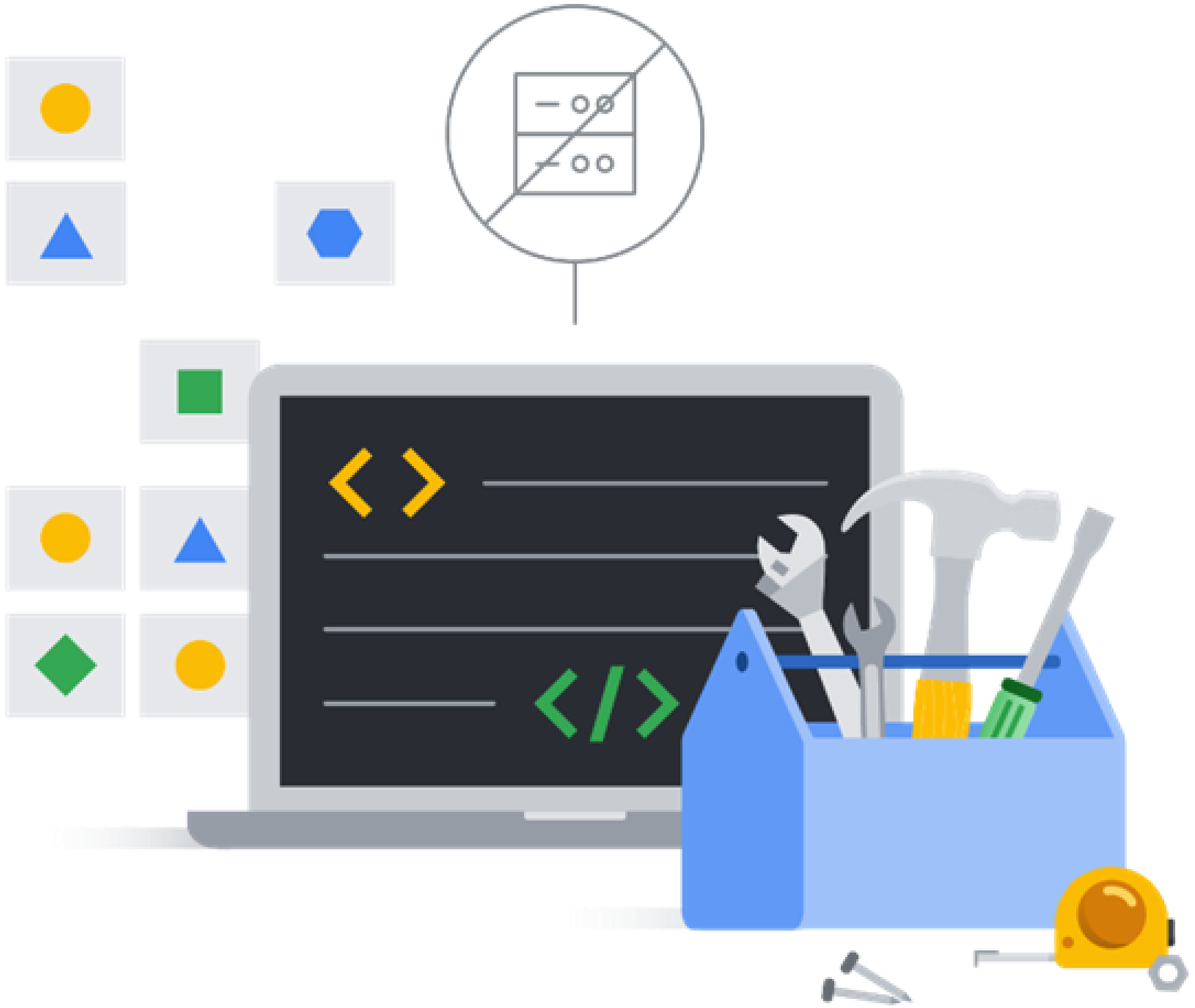
Managed resources

Managed services

Deliver products/services

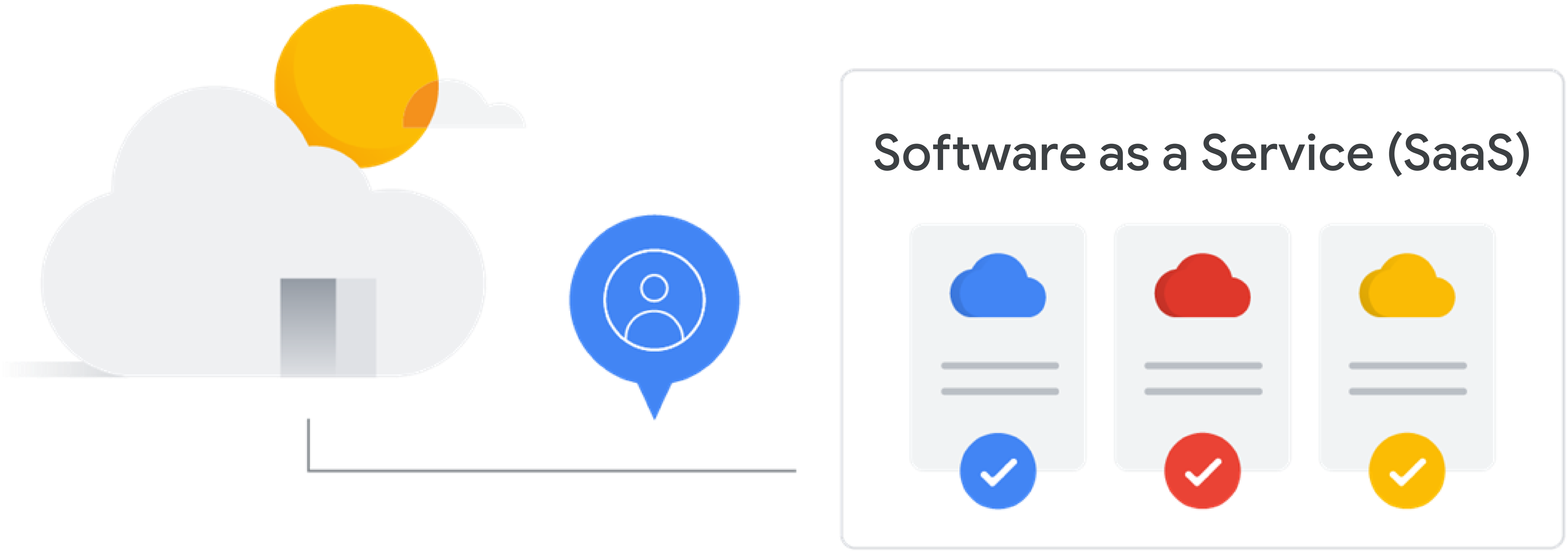
- More quickly
- More reliably

Serverless



- Allows developers to concentrate on code
- No infrastructure management needed

What about SaaS?



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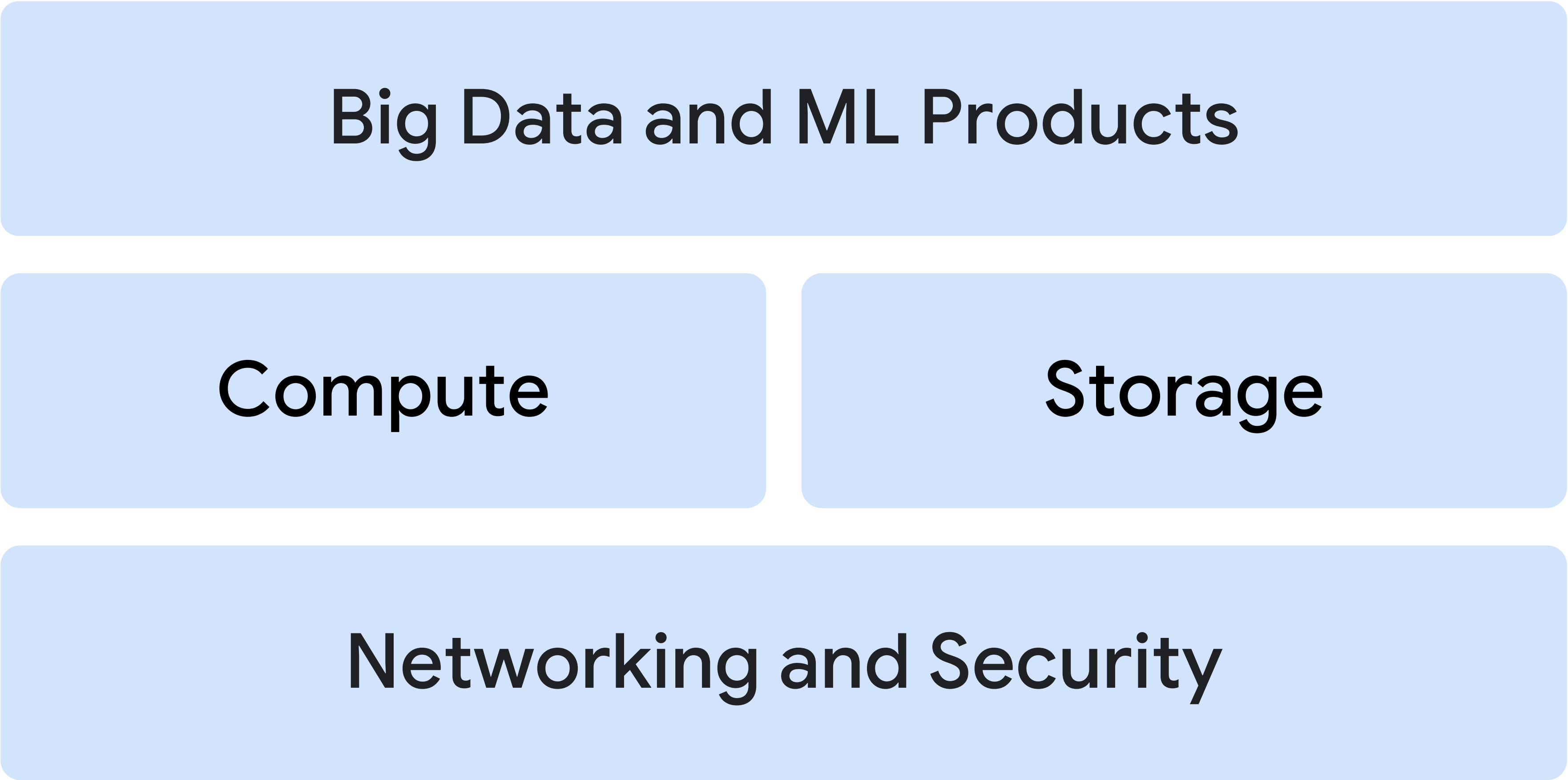
The Google Cloud infrastructure



Google Cloud computing services

Compute Engine

Google Kubernetes Engine

App Engine

Cloud Functions

Cloud Run

Google Cloud storage services

Relational databases

NoSQL databases

Cloud Storage

Cloud SQL

Cloud Spanner

Cloud Bigtable

Firestore

A robust big data and ML product line

Cloud Storage
Dataproc
Cloud Bigtable
BigQuery
Dataflow
Firestore
Pub/Sub
Looker
Cloud Spanner
AutoML
Vertex AI

The Google Cloud infrastructure

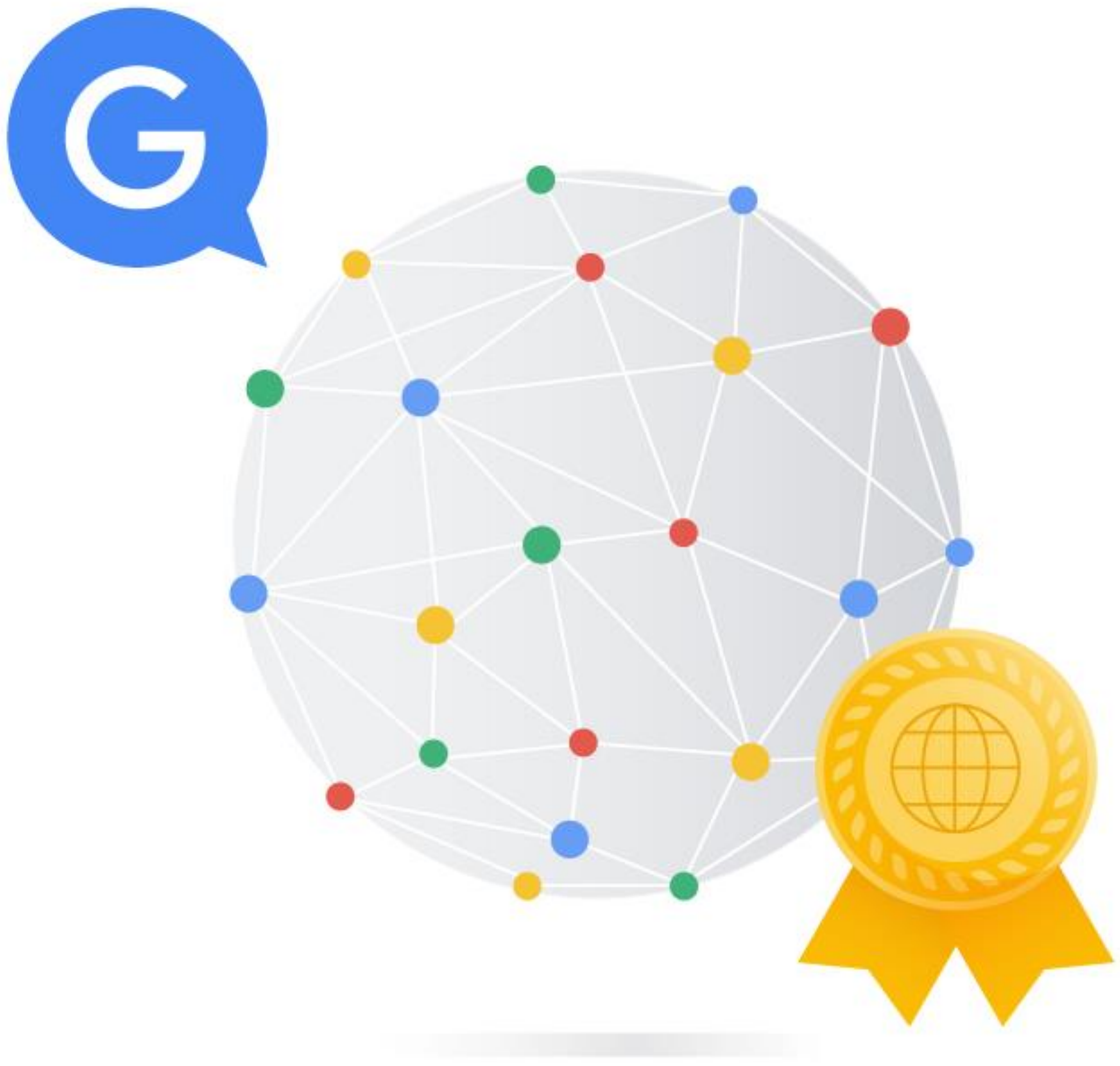
Big Data and ML Products

Compute

Storage

Networking and Security

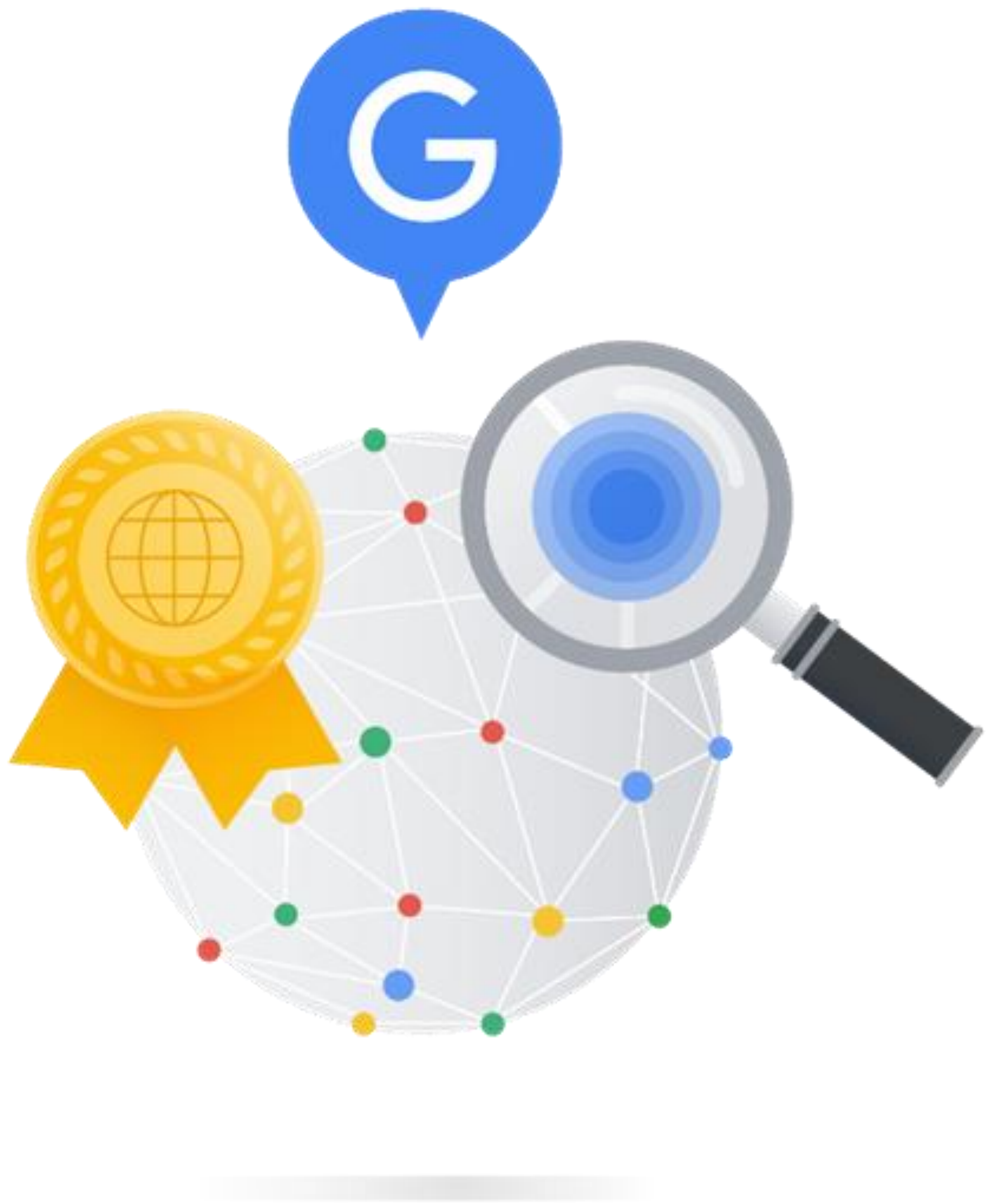
Largest network of its kind



Google's investment in its network runs into the billions of dollars

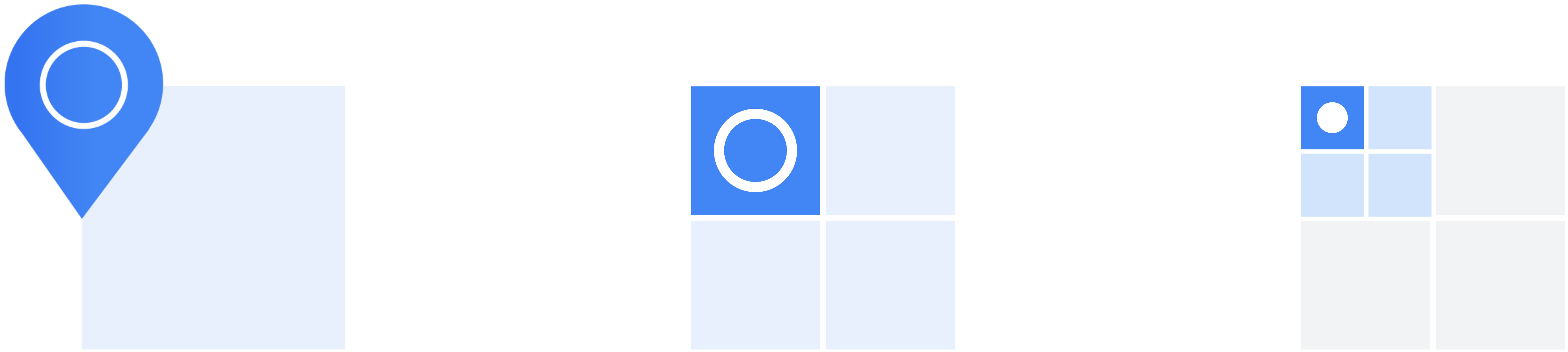
Designed for high throughput

- Highest possible throughput
- Lowest possible latencies
- 100+ content caching nodes worldwide
- High demand content is cached for quicker access



Infrastructure locations

Geographic locations contain regions and zones

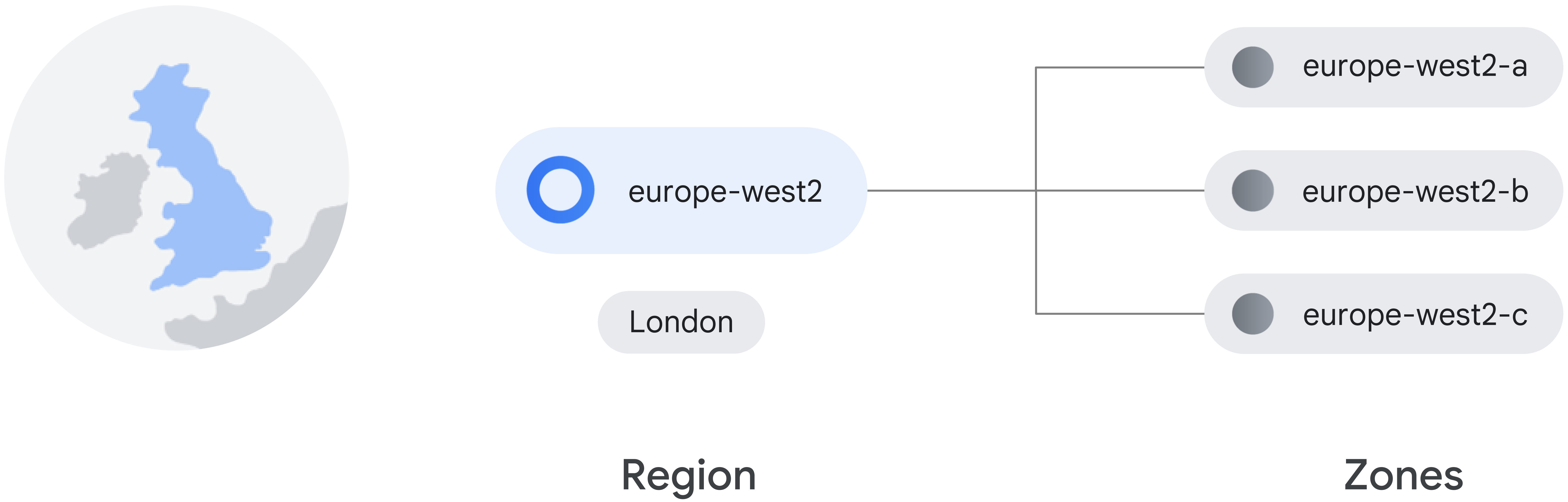


Location

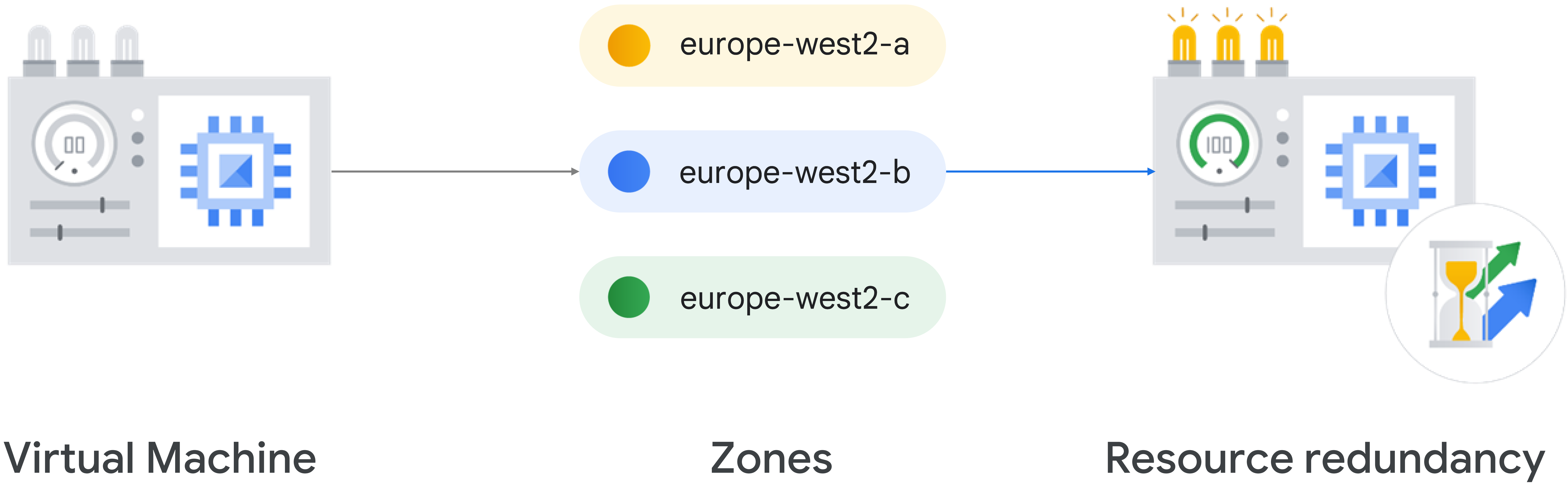
Regions

Zones

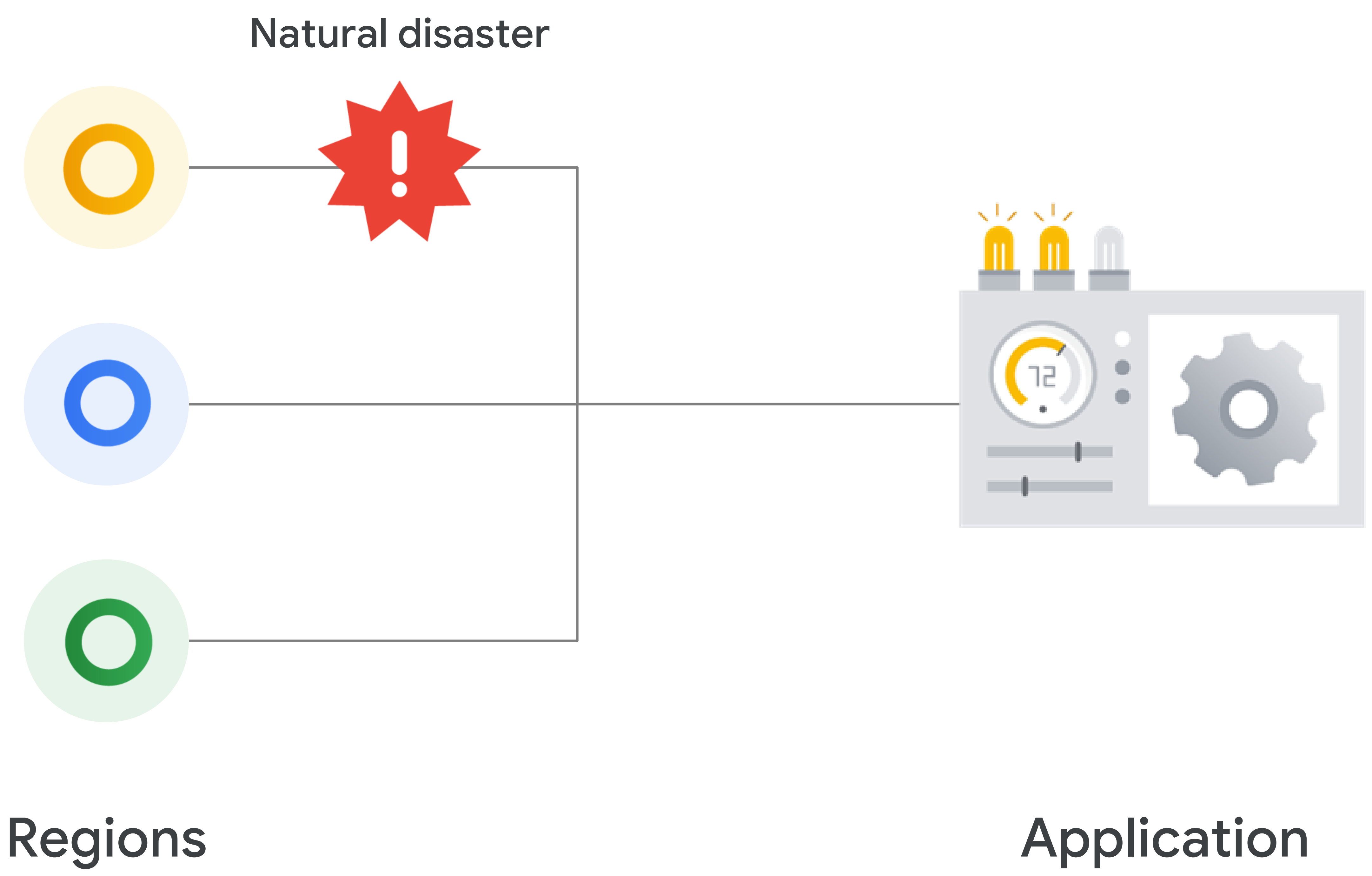
Regions contain multiple zones



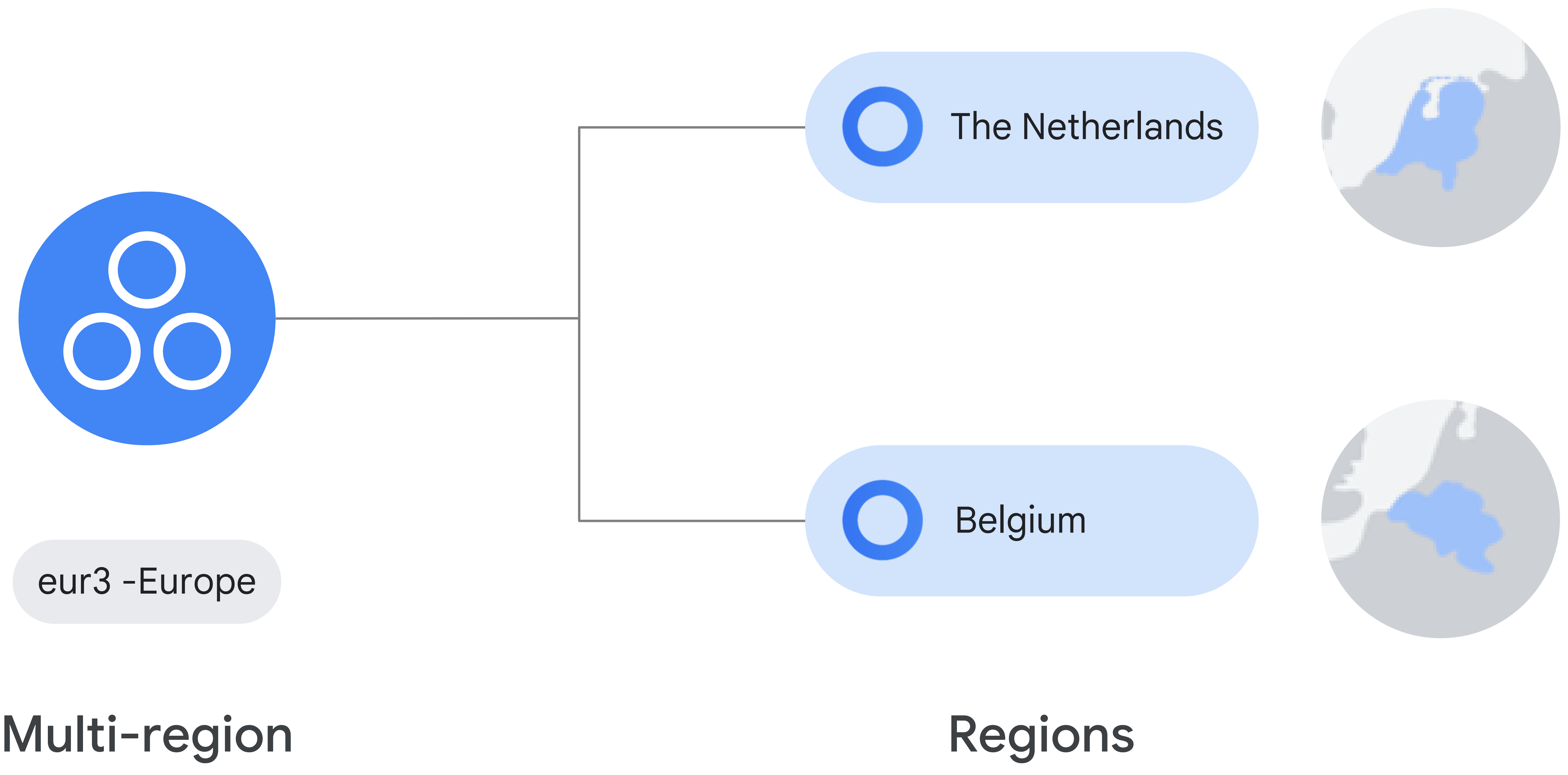
Zones are where Cloud resources are deployed



Resources can run in different regions



Some services can run in multiple geographic locations



103 Zones → 34 Regions

cloud.google.com/about/locations

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Knowledge check

Which is a fundamental attribute of cloud computing?

A. Customers only get computing resources when the cloud provider has availability.

B. Customers get access to computing resources over the internet, from anywhere.

C. Computing resources cannot be scaled up or down.

D. Customers pay for allocated computing resources whether they make use of them or not.

Knowledge check

Which is a fundamental attribute of cloud computing?

A. Customers only get computing resources when the cloud provider has availability.

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Knowledge check

What is the fully automated, elastic third-wave cloud that consists of a combination of automated services and scalable data?

A. On-premises

B. Colocation

C. Virtualized data center

D. Container-based architecture

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Knowledge check

Which service provides raw compute, storage, and network capabilities, organized virtually into resources that are similar to physical data centers?

A. IaaS

B. PaaS

C. SaaS

D. FaaS

Knowledge check

Which service provides raw compute, storage, and network capabilities, organized virtually into resources that are similar to physical data centers?

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C. SaaS

D. FaaS

Knowledge check

Where do Google Cloud resources get deployed?

A. Region

B. Zone

C. Location

D. Multi-region

Knowledge check

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Summary

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- Differentiated IaaS, PaaS, and SaaS.
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- Examined the Google network and how it powers cloud computing.